

MODULE 34 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

The Seven Rs

Each R is chosen by a constraint, not by a preference.

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The question is never which R is best. It is **which constraint the scenario has already given you.**

Three Rs that avoid the work entirely

- ◆ **Retire.** Switch it off. Discovery routinely finds 10 to 20 per cent of an estate with no traffic and no owner.
- ◆ **Retain.** Leave it where it is, deliberately, and revisit later. A conscious decision, not a failure.
- ◆ **Repurchase.** Replace it with a SaaS product. The workload leaves your migration scope completely.
- ◆ Do these first. Every application removed from scope is one you never plan, test, cut over or support.

ON THE EXAM

Retire is the cheapest migration there is. Application Discovery Service finding servers with no network traffic is how you locate them.

Four Rs that move it

STRATEGY	WHAT CHANGES	CHOOSE IT WHEN
Relocate	The hypervisor. Nothing above it	Large VMware estate, exit the data centre fast
Rehost	Where it runs. Not the application	Speed matters; a deadline drives the plan
Replatform	One or two components, managed	A clear win is available for modest effort
Refactor	The architecture itself	Strategically important, and the design is the constraint

ON THE EXAM

Effort and long-term value both rise down this list. Refactoring gives the most and costs the most, in time and money.

The constraint that decides each

- ◆ **A hard deadline** means rehost or relocate. Neither requires application change, so neither depends on developer availability.
- ◆ **A vendor application nobody may modify** means rehost, retain or repurchase. Refactoring is not available to you.
- ◆ **An obvious managed-service equivalent** means replatform. Self-managed SQL Server to RDS is the standard example.
- ◆ **The architecture itself is the problem** means refactor, and only when the application is strategically worth it.

ON THE EXAM

Read for who is allowed to change the code. If nobody is, four of the seven Rs disappear before you compare anything.

Honest notes

- ◆ **Relocate's usual vehicle is VMware Cloud on AWS**, which is now procured through Broadcom rather than the AWS resell channel.
- ◆ **Relocate preserves the operating model**, and therefore the inefficiencies, licensing costs and technical debt that came with it.
- ◆ A typical portfolio lands near **40 to 50 per cent rehost, 20 to 30 per cent replatform, 10 to 15 per cent refactor**.
- ◆ Most portfolios use several Rs. A plan proposing one strategy for everything has not been assessed.

ON THE EXAM

If a proposed plan is 90% refactor, the question is not whether it is technically better. It is whether it will ever finish.

Reading the scenario

THE SCENARIO SAYS

Data centre lease expires in months

No network traffic, no known owner

Self-hosted CRM, email or HR system

Vendor application, no modification allowed

Self-managed database on EC2

The monolith is holding the business back

IT IS POINTING AT

Rehost, or relocate

Retire

Repurchase

Rehost or retain

Replatform to a managed service

Refactor, if it is strategic

ON THE EXAM

Two questions settle most of these: is there a deadline, and is anyone allowed to change the code?

Meridian Freight's data centre lease expires in eight months and will not be renewed. There are 840 servers running 210 applications. The development teams are already committed to a regulatory programme for the next year and cannot take on migration work. Leadership has asked for a modernisation-first approach.

What should the solutions architect recommend?

- A Refactor the applications to cloud-native services during the migration
- B Rehost the majority to meet the deadline, then modernise selected applications afterwards
- C Extend the data centre lease to allow time for modernisation
- D Retain everything on-premises until developers are available

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Refactor during migration
Refactoring needs developers, and the scenario says there are none for a year. Leadership's preference does not create capacity.
- B

 Rehost now, modernise later
Rehosting needs no application change, so it is unblocked by the developer constraint and fits the deadline. Modernisation follows once in AWS.
- C

 Extend the lease
The scenario says it will not be renewed. Changing a fixed constraint is not an architectural recommendation.
- D

 Retain everything
There will be no data centre in eight months. Retain requires somewhere to retain it.

THE SIGNAL

Leadership wants modernisation and the developers are unavailable. When a preference meets a constraint, the constraint wins.

Discovery across 1,100 servers finds 180 with no inbound or outbound network traffic in 90 days, no logged-in users, and no owner listed in the CMDB. Several have hostnames referring to projects that ended years ago.

The migration team has budgeted to migrate all 1,100.

What should the solutions architect recommend?

- A Investigate the 180 for retirement, removing them from scope after a confirmation period
- B Rehost all 1,100, since migrating is safer than deleting
- C Retain the 180 on-premises indefinitely
- D Power off the 180 immediately and migrate the rest

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Investigate for retirement
Retire is the cheapest strategy. Sixteen per cent of the estate leaves scope, with a confirmation period covering quarterly or annual jobs.
- B** Migrate everything
Pays to plan, test, cut over and then run 180 servers nobody uses, indefinitely. Migrating waste preserves it.
- C** Retain indefinitely
Keeps them powered on-premises forever. Retain is for workloads you will revisit, not for ones nobody has identified.
- D** Power off immediately
Ninety days misses quarterly and annual processes. The right instinct, applied without the confirmation period.

THE SIGNAL

Retire is the only R that reduces cost permanently and immediately. Sixteen per cent off the scope is a large result.

A company self-hosts an email platform, a CRM and an HR system on 40 servers. Each requires patching, licensing and a specialist administrator. Mature SaaS equivalents exist for all three, and the business has no requirement that differentiates its usage from any other company's.

The migration team has scoped all three for rehosting.

What should the solutions architect recommend?

- ☐ A Repurchase: move to SaaS equivalents and remove all three from migration scope
- ☐ B Rehost all three as scoped, then evaluate SaaS later
- ☐ C Refactor the three applications to run on containers
- ☐ D Replatform the databases behind each to RDS

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Repurchase to SaaS
Commodity functions with no differentiating requirement, mature products available. Forty servers, three specialists and the licensing all leave scope.
- B** Rehost then evaluate
Migrates forty servers to postpone a decision the scenario has already made for you. Later evaluations rarely happen.
- C** Refactor to containers
Engineering investment in commodity software you did not write. It is the most expensive answer to the least strategic problem.
- D** Replatform the databases
Improves the database layer of applications you should not be operating at all.

THE SIGNAL

No differentiating requirement is the phrase. Building or hosting something that gives you no advantage is the definition of repurchase.

A logistics application runs on Windows Server with a self-managed SQL Server database on EC2 after an earlier lift and shift. The DBA team spends significant time on patching, backups and failover testing. The application uses standard SQL Server features with no unsupported extensions.

There is no deadline pressure and a modest engineering budget.

What should the solutions architect recommend?

- ☐ A Replatform the database to Amazon RDS for SQL Server, leaving the application unchanged
- ☐ B Refactor the application to use Aurora and microservices
- ☐ C Leave it as it is; it already runs on AWS
- ☐ D Repurchase a SaaS logistics platform

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Replatform to RDS for SQL Server

The canonical replatform. Patching, backups and failover become AWS's job, the application does not change, and no unsupported features block it.

C Leave it

Running on EC2 is not the same as being done. The DBA effort the scenario describes continues indefinitely.

B Refactor to Aurora

A far larger project. Aurora is not SQL Server, so this is an engine migration plus a rewrite, on a modest budget.

D Repurchase

Logistics is likely a differentiating capability for a logistics company, unlike the commodity systems in scenario three.

THE SIGNAL

No unsupported extensions is the planted detail. It is the thing that normally blocks a move to a managed engine.

A company runs 400 VMware virtual machines and must exit its data centre in five months. Its infrastructure team is deeply skilled in vSphere and has no AWS experience. Retraining before the deadline is not realistic, and the applications include several the vendor forbids modifying.

Longer term, the company intends to adopt native AWS services.

What should the solutions architect recommend?

- ☐ A Relocate to VMware Cloud on AWS to meet the deadline, then modernise into native services over time
- ☐ B Refactor all 400 workloads to native AWS services now
- ☐ C Retain the estate and negotiate a longer lease
- ☐ D Rehost all 400 with Application Migration Service and retrain the team simultaneously

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Relocate, then modernise
The operating model is unchanged, so the vSphere team stays productive. Five months is achievable, and modernisation follows without a deadline.
- B** Refactor now
Four hundred workloads refactored in five months, by a team with no AWS experience, including applications nobody may modify.
- C** Negotiate the lease
Changes the constraint rather than working within it, and the scenario presents the exit as fixed.
- D** Rehost and retrain
The strong alternative. Rehosting works, and it hands a team with no AWS experience 400 EC2 instances to operate from day one.

THE SIGNAL

The vSphere skills detail is what separates relocate from rehost here. Relocate keeps the team's existing expertise useful.

A retailer's order management monolith cannot scale for peak trading, takes six weeks to release a change, and is the single largest constraint on the business's growth. It is the company's core differentiator. There is executive sponsorship, a dedicated team, and an eighteen-month horizon.

A consultant has proposed rehosting it to AWS first.

What should the solutions architect recommend?

- ☐ A Rehost it first, as proposed, then decide
- ☐ B Refactor it, since the architecture itself is the constraint and the conditions for refactoring are met
- ☐ C Replatform the database and leave the monolith intact
- ☐ D Repurchase a commercial order management product

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Rehost first

Usually right, and here it moves the constraint to AWS unchanged. Six-week releases and no scalability follow it across.

C Replatform the database

Addresses a component that is not the problem. The release cycle and scalability are architectural, not database, issues.

B Refactor

All four conditions are present: the architecture is the constraint, it is a differentiator, and there is sponsorship, a team and time.

D Repurchase

Replaces the company's core differentiator with the same product its competitors use. Sound for commodity, wrong here.

THE SIGNAL

This is the rare case where refactor is right, and the scenario supplies every precondition deliberately. Most scenarios do not.

What to carry out of this module

- ◆ **The constraint chooses the R**, not a preference. Find it before you compare strategies.
- ◆ **Retire, retain and repurchase first**. Every application out of scope is one you never plan, test or cut over.
- ◆ **Refactor needs four things**: the architecture is the constraint, it is strategic, and there is sponsorship and time.
- ◆ **Relocate keeps the operating model**, which is its value and its cost. It defers modernisation rather than answering it.

ON THE EXAM

Two questions settle most of these: is there a deadline, and is anyone allowed to change the code?